Project Report
Storage Device Security Analysis

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Project Proposal

Project Title: Storage Device Security Analysis

Problem Statement

Digital storage devices such as HDDs, SSDs, USB flash drives, and NAS systems are widely used to store sensitive and personal data. Despite their widespread use, most users remain unaware of the security risks associated with improper disposal, unauthorized access, and inadequate encryption of these devices. A simple format operation is commonly mistaken for permanent data removal, leaving sensitive information vulnerable to recovery by malicious actors. There is a clear need for a structured approach that addresses storage device security across its full lifecycle.

Proposed Solution

This project proposes a three-phase security framework for storage devices:

- Prevention: Implement AES-256 full-volume encryption on removable media using VeraCrypt to ensure data confidentiality in the event of theft or loss.
- Detection: Deploy a centralized monitoring system using Wazuh SIEM to log and alert on USB connection events and file transfer activities in real time.
- Response: Establish a forensically sound investigation procedure using FTK Imager, SIFT Workstation, Autopsy, and supporting tools to acquire, analyze, and recover data from suspect storage devices without compromising evidence integrity.

Expected Outcomes

- A verified forensic disk image of a USB flash drive with confirmed hash integrity.
- Successful recovery of deleted and formatted data including documents, images, and email artefacts.
- A complete activity timeline reconstructed from the forensic image.
- A structured forensic report documenting the full chain of custody and investigation findings.
- Demonstrated proof that quick formatting does not permanently erase data, highlighting the need for proper sanitization practices.

Scope and Limitations

The project focuses on removable USB storage media as the primary test device. The analysis will be conducted in a controlled lab environment using VMware virtualization. Network-attached storage and cloud forensics are referenced in the system architecture but are outside the primary scope of practical implementation. The project does not involve any real-world evidence or live criminal investigation.

Abstract

This project presents a comprehensive security analysis of digital storage devices, focusing on the prevention, detection, and forensic investigation of data breaches involving removable media. The study demonstrates a complete forensic workflow applied to a USB flash drive, beginning with AES-256 encryption using VeraCrypt for data protection, real-time USB event monitoring through Wazuh SIEM for threat detection, and forensically sound evidence acquisition using FTK Imager with SHA-256 and MD5 hash verification. The acquired disk image was analyzed within the SIFT Workstation environment using industry-standard tools including Autopsy, The Sleuth Kit, Bulk Extractor, Scalpel, and PhotoRec to recover deleted files, extract digital artefacts, and reconstruct activity timelines. Results confirmed that despite a quick format operation, all deleted data was successfully recovered, including sensitive files and over 10,000 email-related artefacts from an apparently empty device. This investigation demonstrates that quick formatting provides no meaningful data protection, and that open-source forensic tools are fully capable of conducting professional-grade digital investigations. The findings emphasize the critical importance of cryptographic erasure or physical destruction when disposing of storage media containing sensitive information.

1. Introduction

The rapid increase in use of digital storage devices in computing environment has introduced significant challenges in the field of cybersecurity and digital forensics. Storage devices including HDD, SSD, USB flash drives and NAS (Network-Attached storage) serve as primary repository for sensitive and personal data.

- **The Scope:** The system covers the entire security lifecycle: **Prevention** (encryption), Detection (live monitoring), and **Response** (digital forensics).

This report presents a comprehensive security analysis of storage devices, examining vulnerabilities, data persistence behaviors, and the forensic methodologies employed to recover, analyze, and authenticate digital evidence. This project was conducted using an industry-standard forensic toolkit

2. Objective

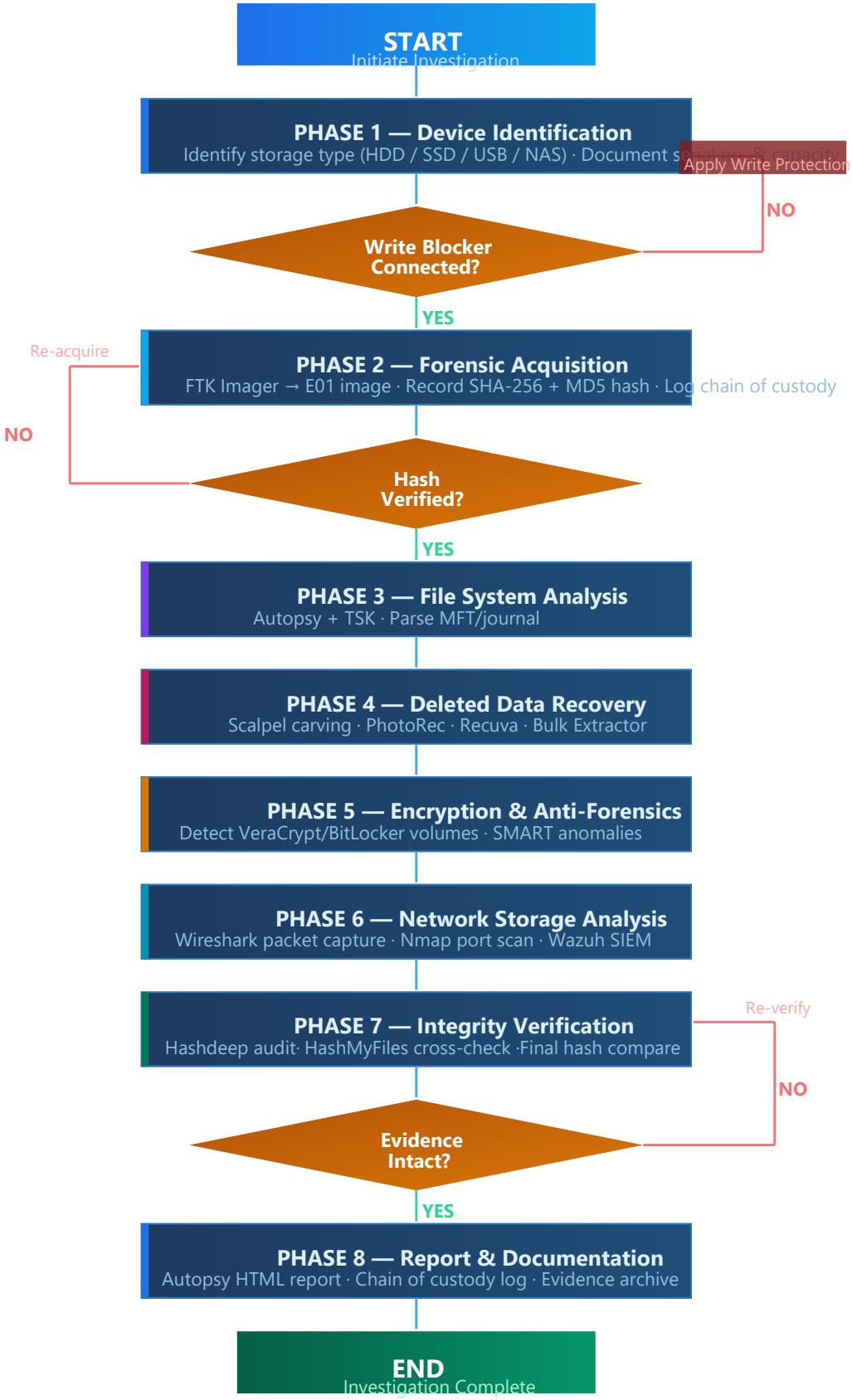
- **Prevention:** Enforce data confidentiality on removable media using AES-256 full-volume encryption (VeraCrypt).
- **Detection):** Implement a centralized monitoring system (Wazuh) to capture, log, and alert on physical USB connection events and file transfer activities in real-time.
- **Evidence Preservation):** Establish a forensically sound procedure to acquire bit-by-bit images (FTK Imager) of suspect storage devices without altering the original data.
- **Forensic Analysis):** Perform post-incident investigations (using Autopsy or SIFT) to recover deleted files, analyze metadata, and reconstruct the timeline of a compromised device.

3. Tools and Technologies

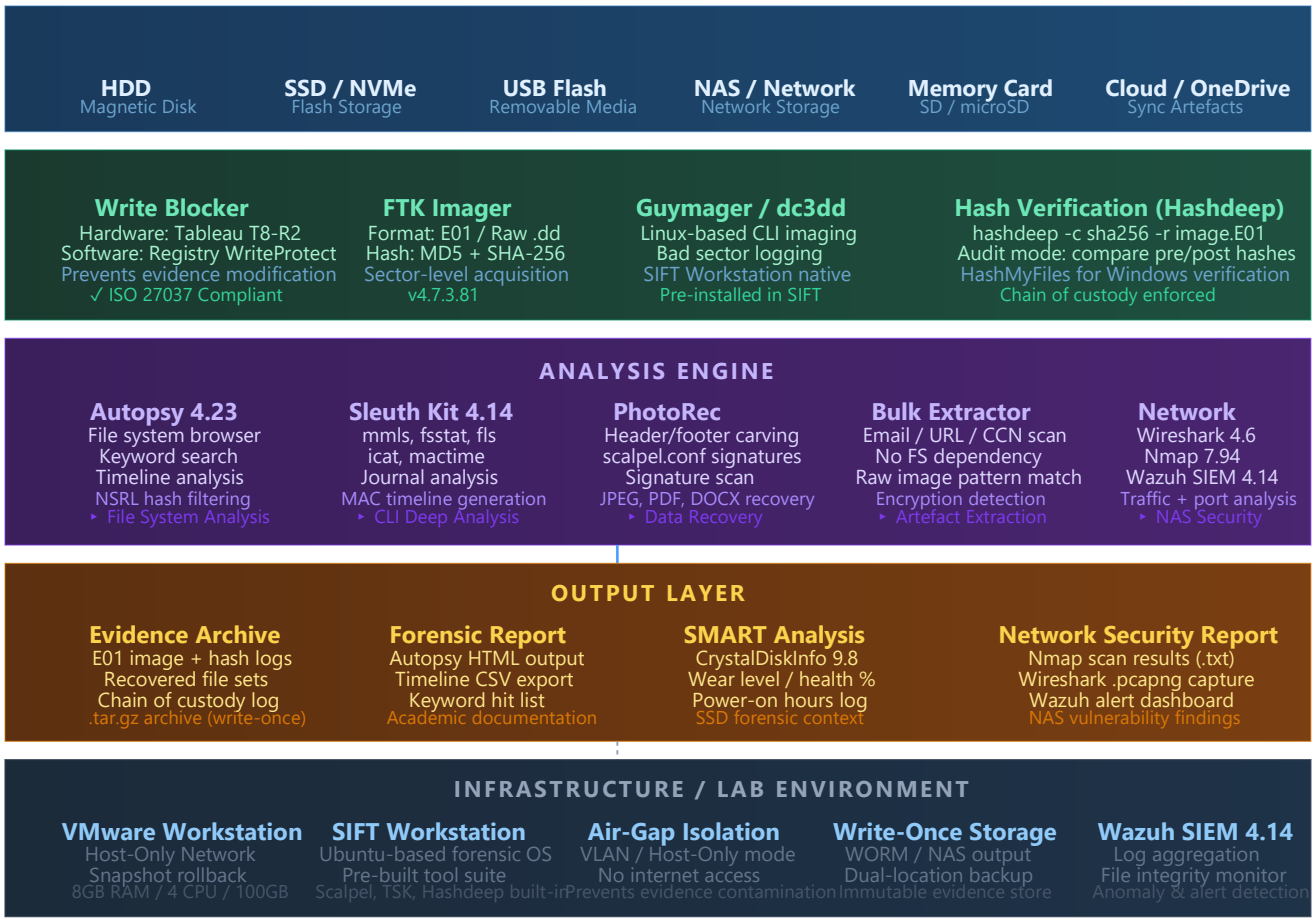
- VeraCrypt
- Exterro FTK Imager
- HashMyFiles
- CrystalDiskInfo, TestDisk
- Autopsy
- Sleuthkit
- WireShark
- Recuva
- SIFT WorkStation
- VMWare

4. Design Phase

4.1 System Flow/Flowchart



4.2 System Architecture



5. Methodology

5.1 Evidence Identification

Each storage device was identified, catalogued, and documented prior to any interaction. Device type, capacity, serial number, and connection interface were recorded as part of the initial case log.

5.2 Forensic Acquisition

Write protection was enforced before any device was connected to the analysis workstation. Forensic images were acquired using FTK Imager in Expert Witness Format (E01), with concurrent MD5 and SHA-256 hash generation to establish a cryptographic baseline for evidence integrity.

5.3 File System Analysis

Acquired images were examined using Autopsy and The Sleuth Kit to parse file system structures, recover deleted entries, and generate MAC (Modified, Accessed, Changed) timelines. Keyword searches and hash filtering against the NSRL database were applied to identify relevant files and eliminate known-good system files.

5.4 Data Recovery

Deleted and fragmented data was recovered using a layered carving approach. Scalpel was configured with file-type signatures for targeted recovery, while PhotoRec provided broader format coverage. Bulk Extractor was used to extract embedded artefacts including email addresses, URLs, and encryption indicators from raw image data.

5.5 Integrity Verification

Hash audits were performed at each stage transition using Hashdeep in audit mode, comparing pre- and post-analysis hash sets to confirm no unintended modification had occurred. All findings were documented contemporaneously in a structured chain of custody log.

5.6 Reporting

Results were compiled into a structured forensic report exported from Autopsy, supplemented by timeline CSVs. The complete evidence archive was stored on write-once media with dual-location backup.

6. Implementation

6.1 Disk image Creation

We had to set a base for our demonstration, for that purpose I have used a USB drive and save some sample data in it, then deleted and furthermore formatted the USB drive.

Next, we have to enable write protection so that no changes occur to our sample USB. This is done by running this command in cmd as administrator

```
reg add "HKLM\SYSTEM\CurrentControlSet\Control\StorageDevicePolicies" /v WriteProtect /t REG_DWORD /d 1 /f
```

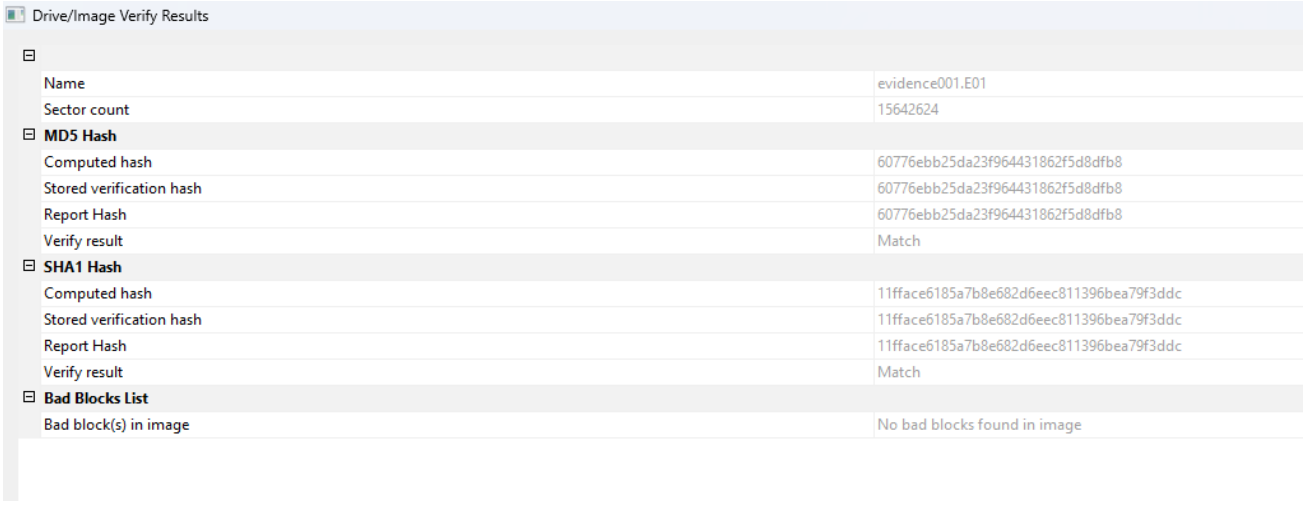
Then we would create disk image using FTK Imager by running as Administrator

- Go to File → Create Disk Image
- Select Physical Drive → click Next
- From the dropdown select your USB drive
- Click Finish

Set Image Destination

- Click Add
- Image type: E01
- Click Next and fill in required fields
- Click Next
- Click Finish

After completely making disk image FTK Imager will give us HASHES



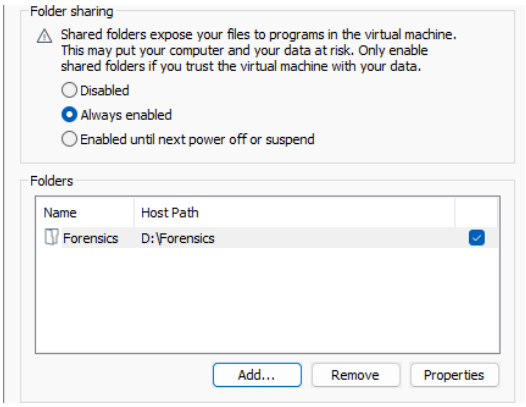
We have to keep them save and compare later to check integrity.
Now disable write protection by

```
reg add "HKLM\SYSTEM\CurrentControlSet\Control\StorageDevicePolicies" /v WriteProtect /t REG_DWORD /d 0 /f
```

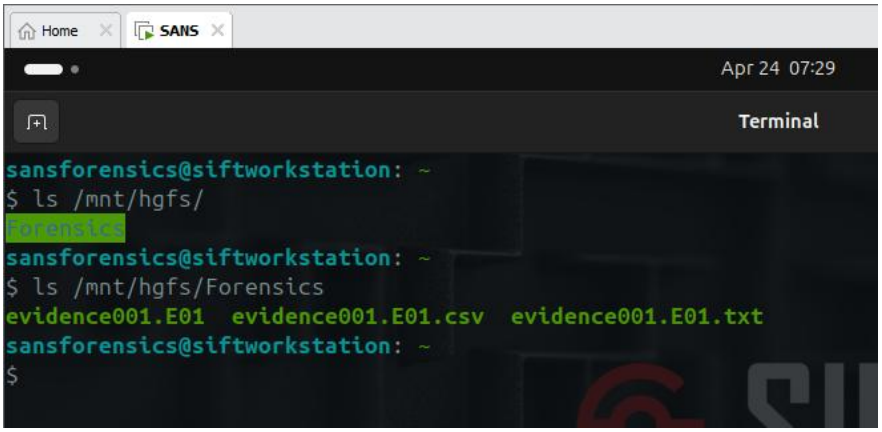
6.2 Working in SIFT

6.2.1 Create Shared folder

Next step is to verify the disk image that is created in above steps
Setup a shared folder with VMWare SIFT



It is visible in SIFT



Now compare hashes, they should be matched

6.2.2 Examine image structure

Find partition table of disk created from USB, using mmls



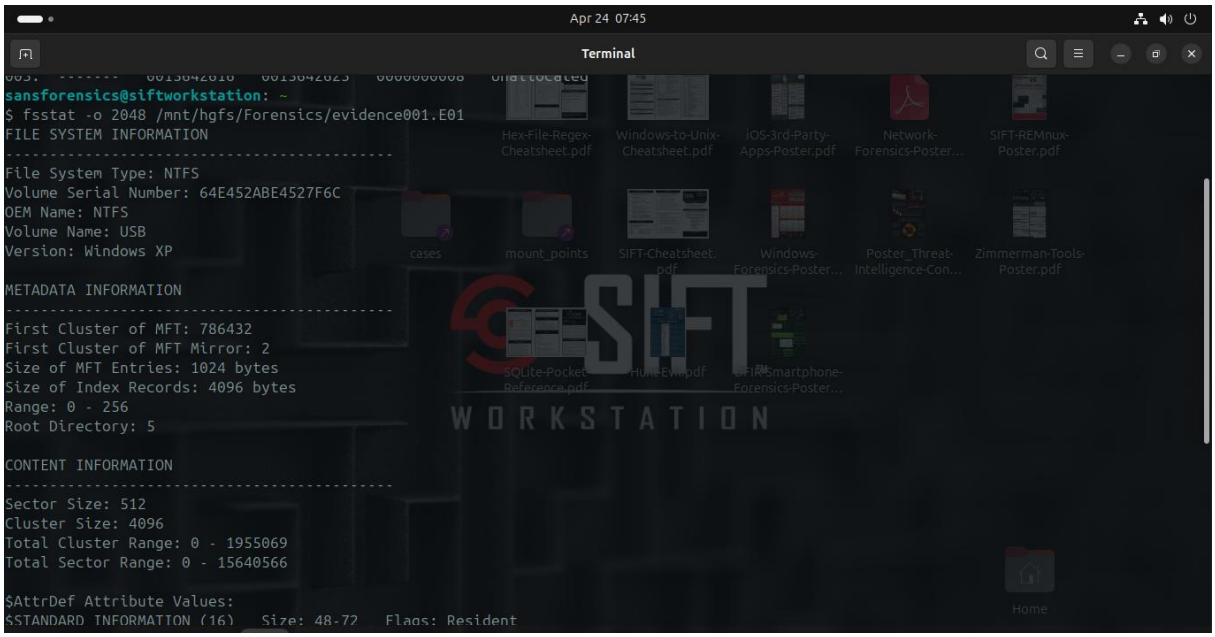
6.2.3 Inspect the file system

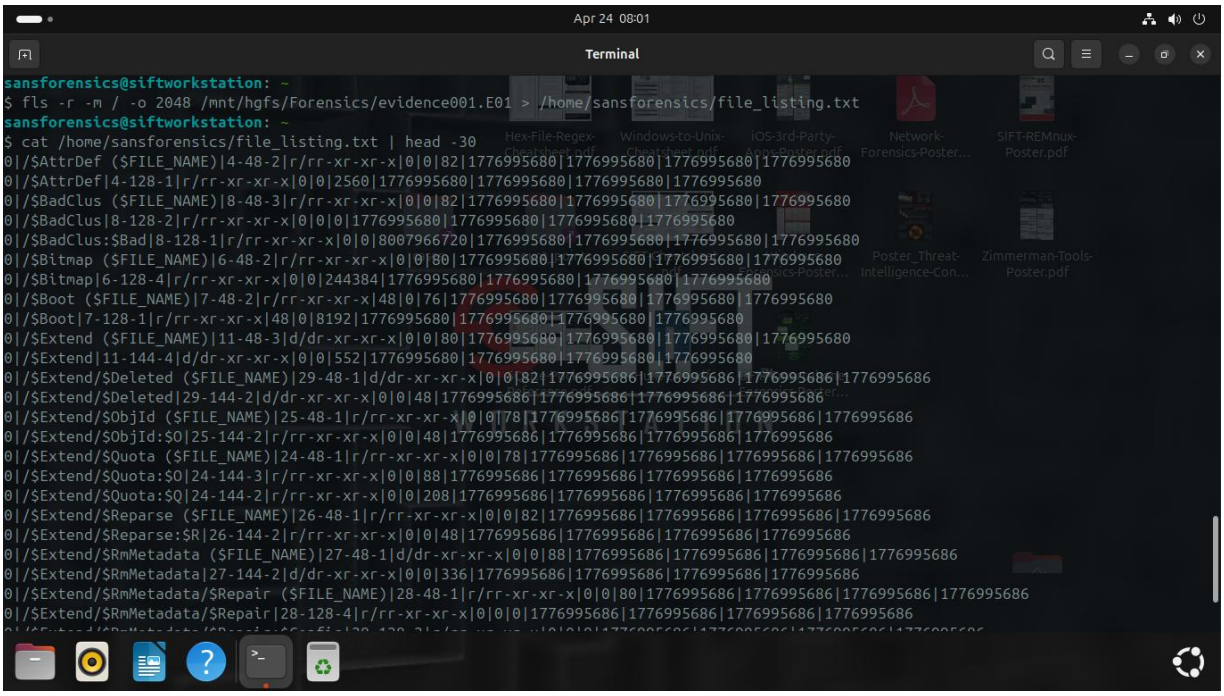
Further, inspect files system to get more detail about partition, file system, cluster size, serial number, Oss and metadata
After that run these commands to get file and folder list, including deleted ones

```
fls -r -m / -o 2048 /mnt/hgfs/Forensics/evidence001.E01 >
/home/sansforensics/file_listing.txt
```

Open using

```
cat /home/sansforensics/file_listing.txt | head -30
```



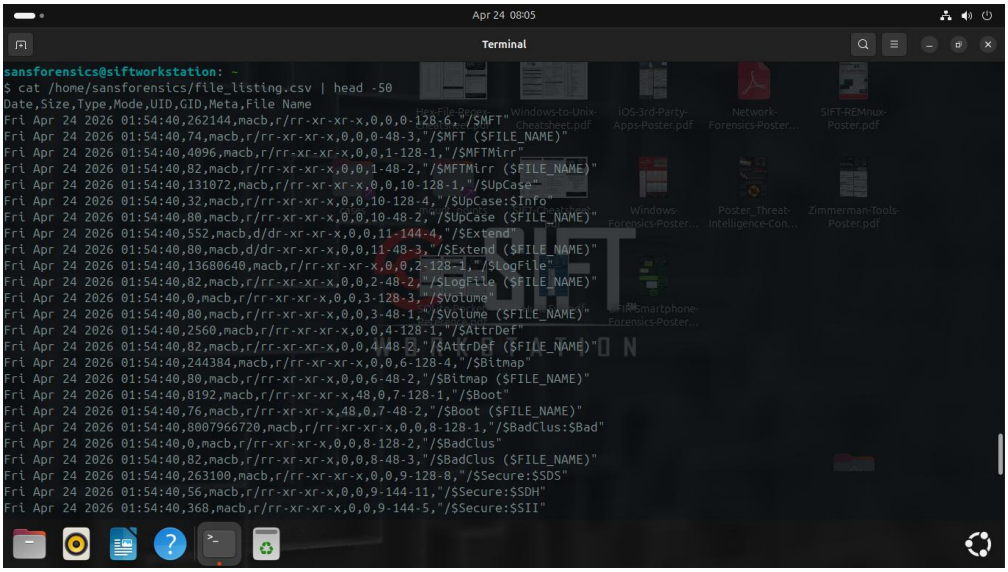


6.2.4 Convert to CSV

Convert .txt in csv format

```
mactime -b /home/sansforensics/file_listing.txt -d > /home/sansforensics/timeline.csv
```

Data enlisted in lising.csv

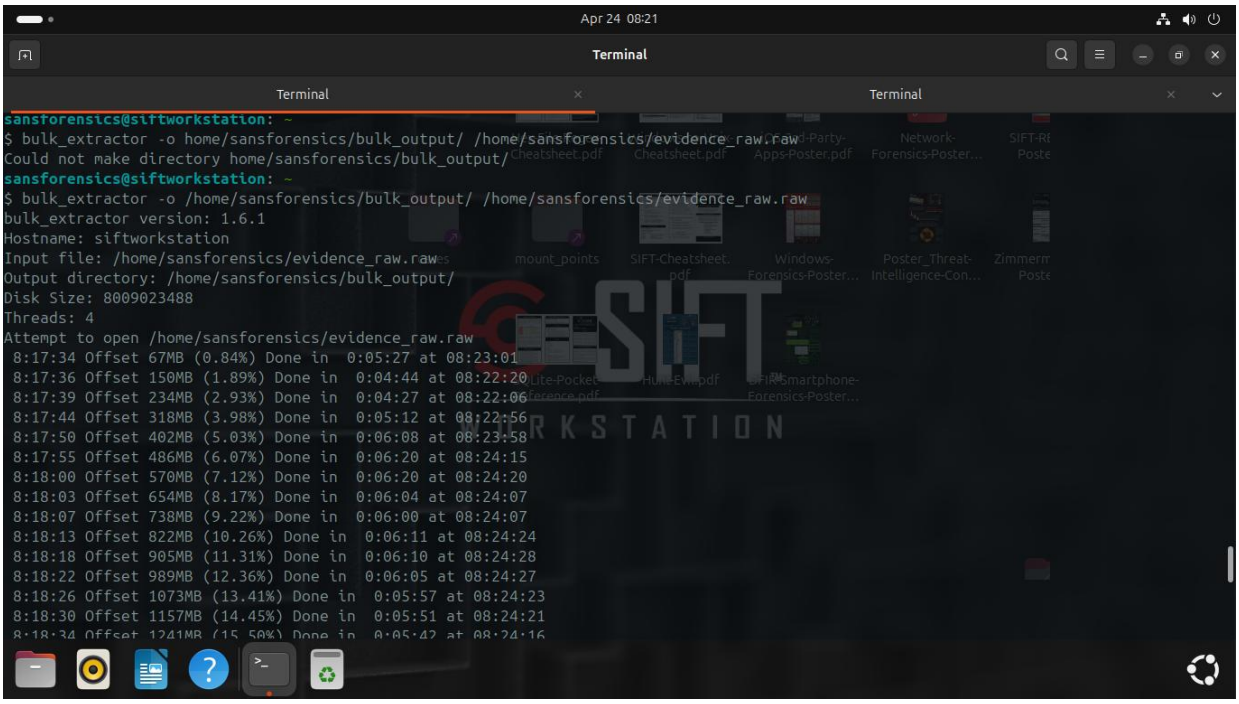


6.2.5 Extraction of Information

Now extract information using bulk Extractor, firstly change the format to set the stage for bulk Extractor

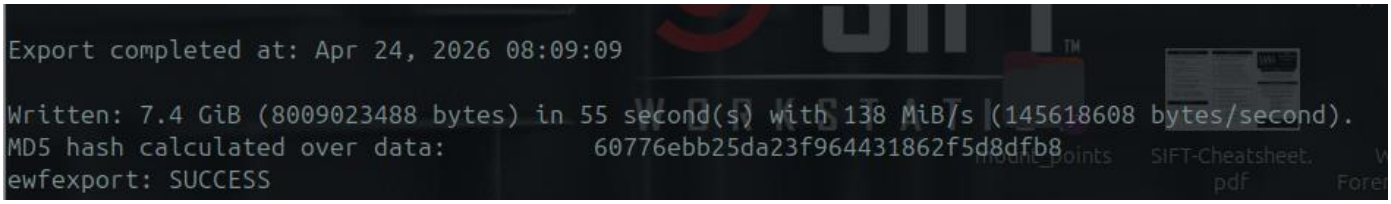
```
ewfexport -t /home/sansforensics/evidence_raw /mnt/hgfs/Forensics/evidence001.E01

now,
bulk_extractor -o /home/sansforensics/bulk_output/ /home/sansforensics/evidence_raw.raw
```



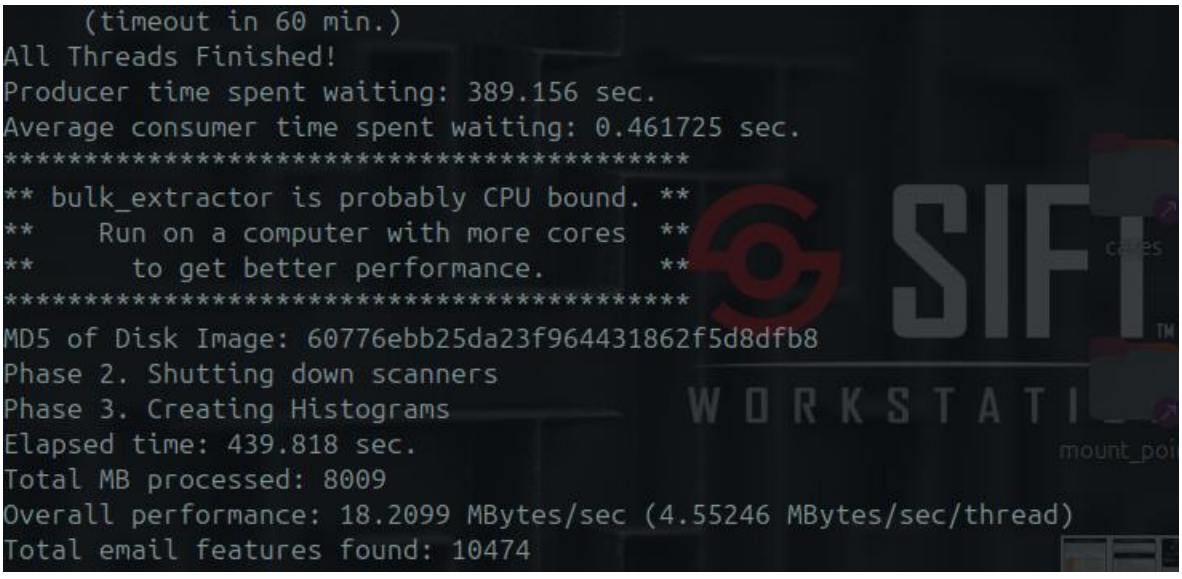
6.2.6 Hashes Matching

Also, the hash is same as FTK has provided

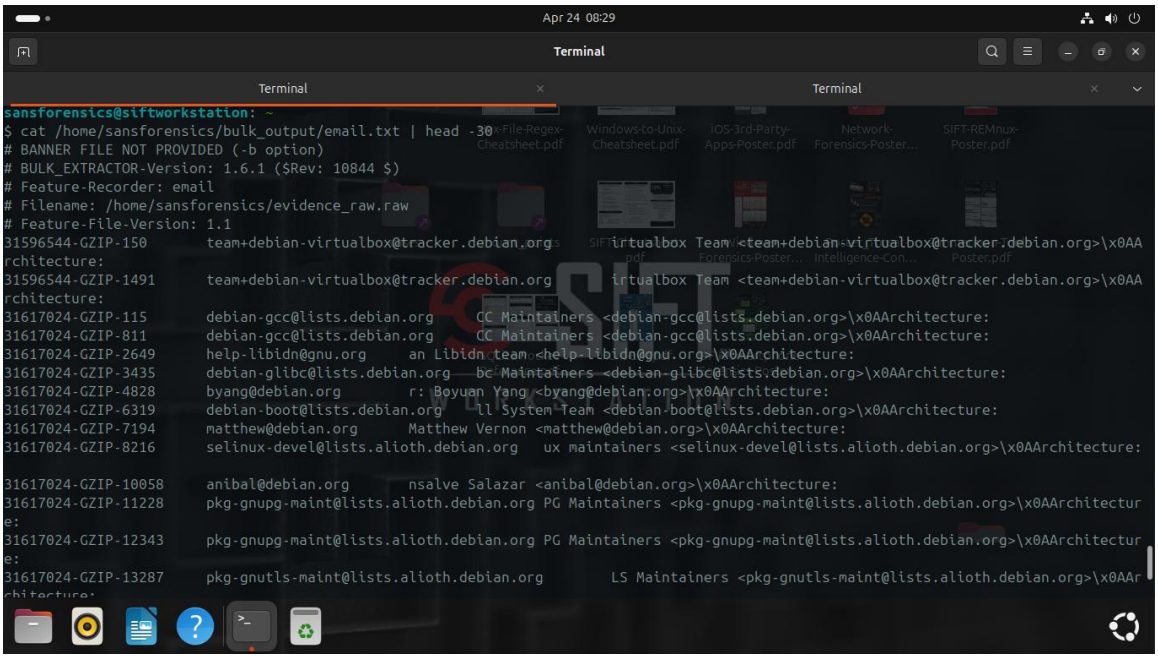


6.2.7 Findings

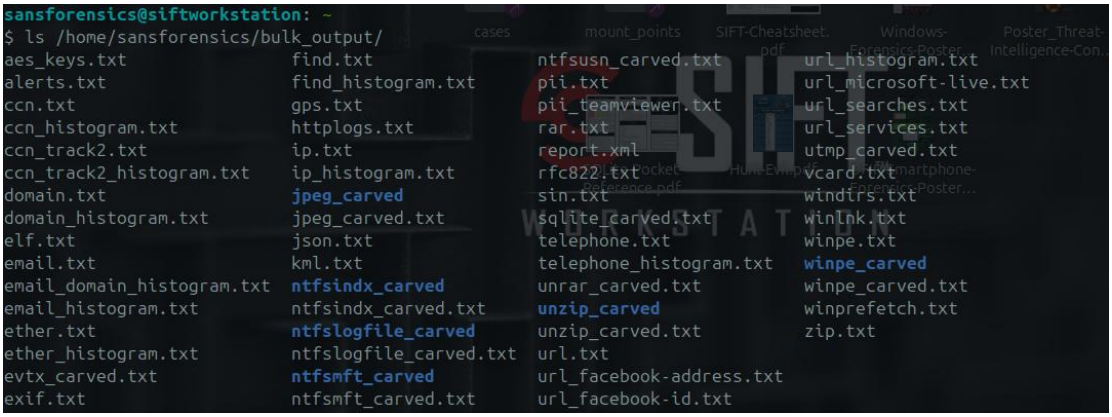
This is the list of what we found



cat /home/sansforensics/bulk_output/email.txt | head -30



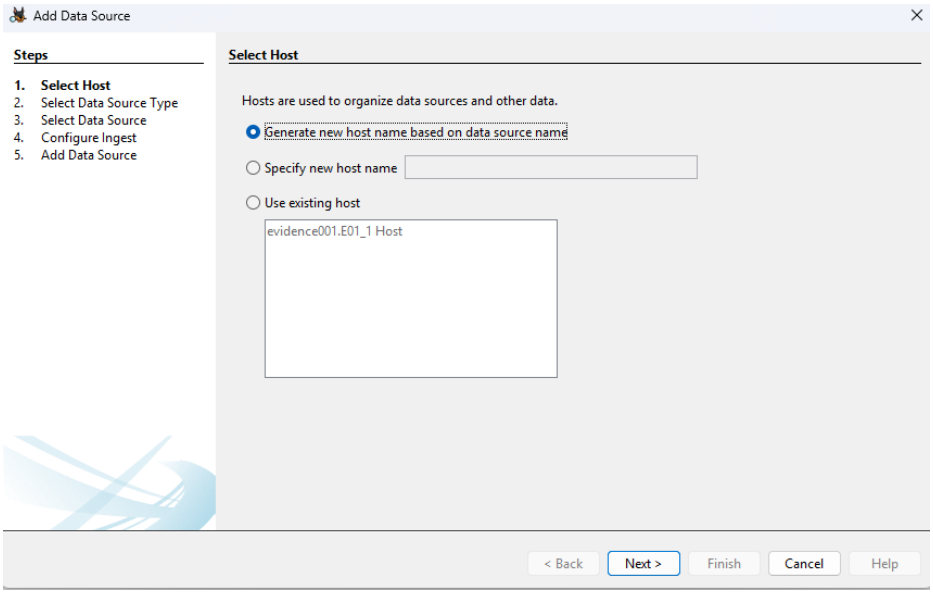
List of files bulk_output



6.3 Working on Autopsy

6.3.1 Add Source data

Open Autopsy as administrator, wait until loading, select new case, add data source

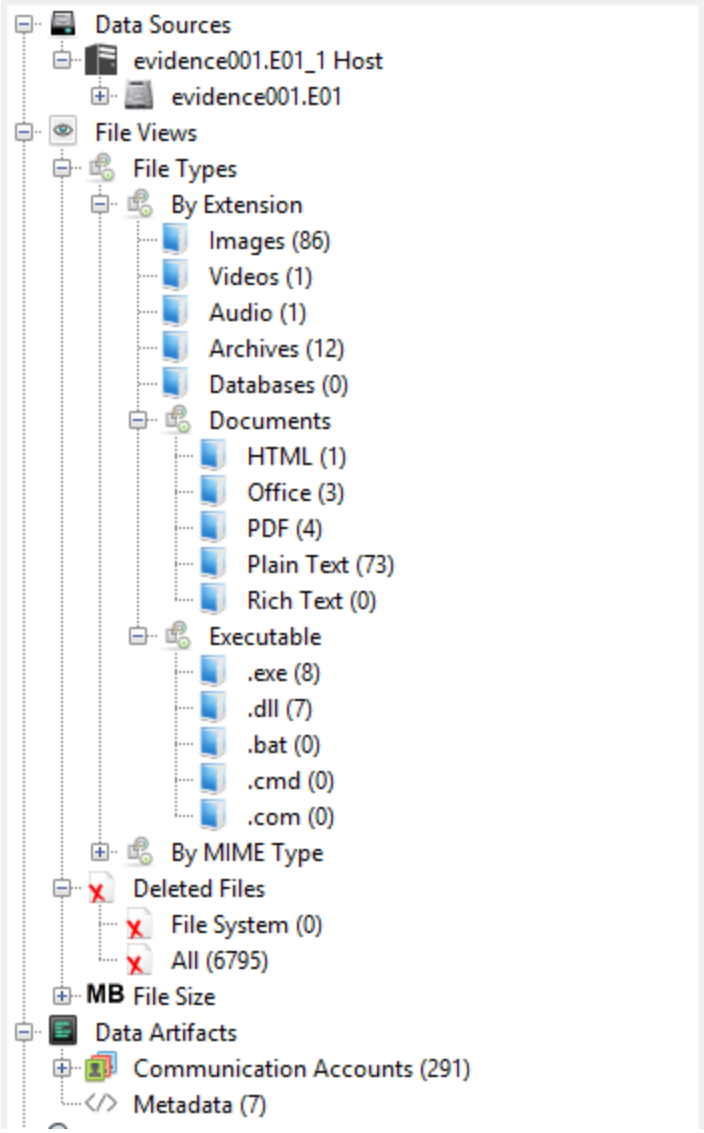


Steps to follow:

- Select data source type as disk image or VM file, click next
- Give path to source file, click next
- Select the required modules according to your need and click next again
- Scanning would be started and take time

6.3.2 Explore Result files

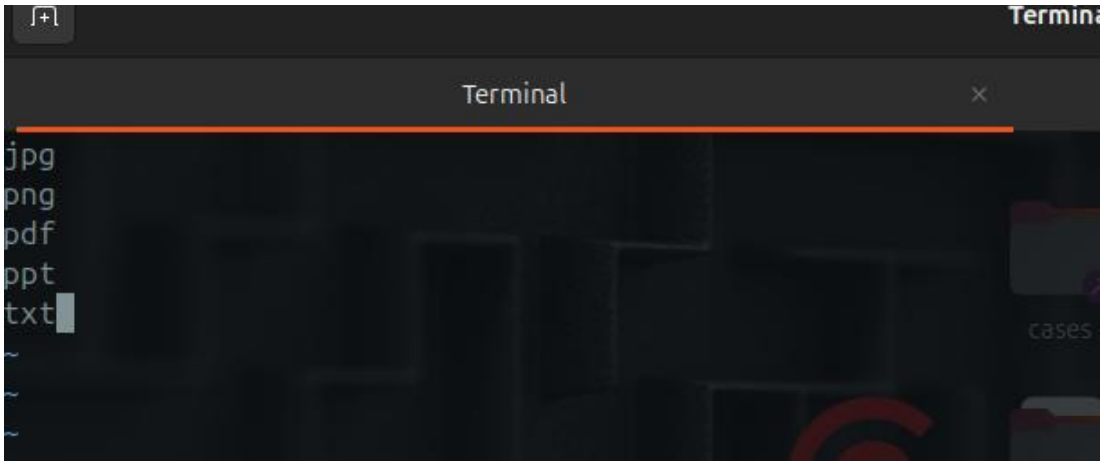
- On left panel, expand data sources → Evidence001.e01
- Navigate through panel and explore files



6.4 Using scalpel

In SIFT terminal, run

- Vi /etc/scalpel/scalpel.conf
- Provide required extentions



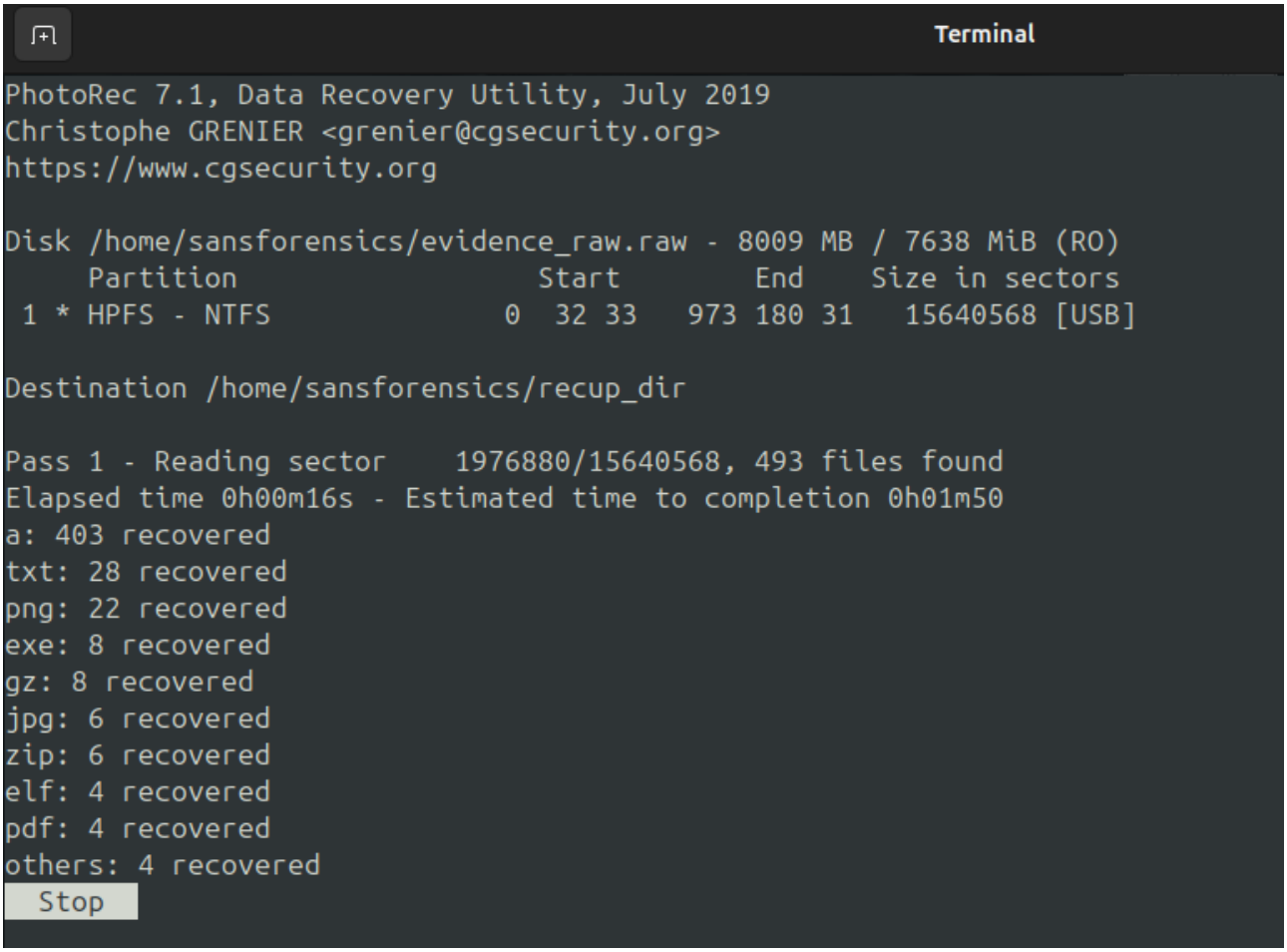
- Press esc when done writing

```
Scalpel /home/sansforensics/evidence_raw.raw -o /home/sansforensics/scalpel_output
```

6.5 PhotoRec

Now with photorec

```
Sudo photorec /home/sansforensics/evidence_raw.raw
```



7. Result

A USB flash drive was forensically imaged using FTK Imager. Hash verification confirmed the image was an exact copy of the original device with MD5 60776ebb25da23f964431862f5d8dfb8 matching across all three verification points FTK Imager, ewfexport, and Bulk Extractor.

Despite a quick format being performed before acquisition, all deleted files were successfully recovered including passwords_backup.txt, bank_statement.txt, and deleted images. Bulk Extractor processed 8,009 MB of raw image data and extracted 10,474 email-related artefacts from a device that appeared completely empty.

8. Conclusion

This project proved that deleted data is never truly gone. A USB drive that appeared fully wiped after formatting was completely reconstructed deleted files recovered, sensitive artefacts extracted, and a full activity timeline rebuilt.

The open-source forensic toolkit used FTK Imager, SIFT Workstation, Autopsy, and Bulk Extractor successfully handled every phase of the investigation at no cost, proving it is fully viable for academic forensic research.

The key security finding is that quick formatting provides no real data protection. Anyone disposing of a storage device without cryptographic erasure or physical destruction risks serious data exposure.

Reference

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